How Far Can One Run of SHAP Be Trusted? Repeat-Run Reliability of Sampling-Based Attributions on Tabular Models

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Abstract

SHAP attributions for tabular models are usually computed by a randomized estimator under a fixed compute budget, and often from a single run. We ask when that run can stand in for "the" attribution of a prediction, and separate four notions of reliability: run-to-run repeatability, accuracy against the exact Shapley vector, resolvability of the feature ordering against the run-to-run noise, and the size of the explainer noise next to background-sample and model-refit variation. The evidence covers two stochastic datasets (German Credit, d=20; ACSEmployment, d=16), one background size (50 rows), one fit per model, one library version, and budgets from $1/4\times$ to $4\times$ the library default (50 instances and 20 runs per instance; $16\times$ and the exact target of the German Credit MLP use 20 instances), with two low-dimensional controls where the default is deterministic. With ℓ_1 selection off, a single default-budget Kernel SHAP run recovers the exact top-3 feature set in at least 0.98 of cases for all four models, and meets a pre-stated reliability rule in all of them; for the MLP, the only model here for which sampling is the practical option, the top-3 hit rate is 0.98 (German Credit, 20 instances) and 1.00 (ACSEmployment). The conditions are narrower than that: with the library's default ℓ_1 setting, a configuration pitfall of that implementation, the rule is not met on German Credit (relative ℓ_2 error 0.12–0.25, a bias repeated runs do not remove); a single run reproduces the order of an adjacent feature pair with probability 0.94–0.95, but only 0.69–0.77 of adjacent pairs are separated by twice the run-to-run noise; and some instances are less repeatable than the average. At the 50-row background, redrawing the background changes the attribution vector far more than the explainer seed (variance ratio roughly 81–150 and 840–1100), which is a comparison of different kinds of variation, not of accuracy. The results say nothing about larger d or other background sizes.

1 Introduction

Shapley-value attributions [Lundberg and Lee, 2017] are used to justify individual predictions of tabular models in credit, health and similar settings. Outside tree models and linear models, where exact algorithms exist [Lundberg et al., 2020], the attribution is *estimated* by sampling coalitions or permutations, with a budget of model evaluations that the practitioner fixes. A common practice is to run the explainer once for an instance and report the resulting vector as the reason for that prediction. If the estimator is noisy, a second run could report a different top feature, a different sign or a different order, and the reported "reason" would then be partly an artefact of the random seed.

Prior work quantifies how estimator variance falls with the number of samples [Covert and Lee, 2021, Mitchell et al., 2022, Musco and Witter, 2025], how explanations differ between methods [Krishna

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et al., 2024], and how the background set matters [Yuan et al., 2023, Chen et al., 2020]. What a practitioner needs is different: for a trained model, a given instance and a realistic budget, *how far can one run be treated as the attribution of that prediction, and where does that fail?* Answering this requires deciding what "reliable" means, against which reference, and under which conditions; a single agreement number between two runs is not enough, because it cannot tell noise from bias and cannot say whether the noise matters for what a reader would conclude.

We study this question empirically with four separable notions of reliability (Section 3): repeatability between independent runs, accuracy against the exact Shapley vector for the same model, background and value function, resolvability of adjacent features against the run-to-run noise, and the size of explainer noise relative to other variation that a deployed explanation already carries. We evaluate them in the regime where sampling noise exists at all: Kernel SHAP enumerates every coalition once the number of samples reaches $2^d - 2$, so for $d \leq 10$ the library default is deterministic; our core datasets therefore have $d = 20$ and $d = 16$, and two lower-dimensional datasets serve as controls.

Findings (scope: $d \leq 20$, background of 50 rows, one fit per model). (i) With the library's default sample count and ℓ_1 selection switched off, a single Kernel SHAP run recovers the top-3 features with mean hit rate between 0.977 and 1.000 on German Credit and 0.999 and 1.000 on ACSEmployment, and the decision rule classes all eight cells as reliable. Two of the four models are not informative about sampling noise (LR is additive, so the estimate is exact; XGBoost and RF have an exact algorithm that one would use instead), so we also state the MLP alone: hit rate 0.977 (20 instances) and 1.000. Permutation SHAP behaves similarly. (ii) The library's default ℓ_1 setting is the one configuration in which the rule is not met ("conditional") on German Credit; the error is a deterministic bias that repeating runs does not reduce. This is a configuration pitfall of one implementation. (iii) Reliability is not uniform inside an attribution vector: a single run reproduces the order of an adjacent pair with probability 0.935–0.947 on German Credit, and 0.695–0.770 of adjacent pairs are separated by twice the run-to-run noise, with the unresolved pairs at the lower ranks. (iv) At a background of 50 rows, redrawing the background changes the attribution vector with variance 81–152 (German Credit) and 843–1102 (ACSEmployment) times that of the explainer seed; since a redraw changes the target rather than the estimate, this describes sizes of variation, not accuracy. Section 5 states what these results do not cover.

Contributions. (1) An evaluation design that combines standard stability and error measures (and a decision rule fixed in the study plan before the runs) with exact targets, for the single-run practice at the library's default budget; the metrics themselves are not new. (2) A controlled evaluation on two stochastic and two deterministic-regime datasets, four model families and two sampling explainers, with nested-bootstrap uncertainty and exact targets for every explained model. (3) A measured configuration pitfall: the default ℓ_1 setting of `shap` 0.51.0's `KernelExplainer` biases single-run attributions at the default budget on a $d = 20$ dataset (the arm was run at $1\times$ only), and averaging runs at equal total budget does not improve on one larger run in our data (a small effect). (4) All code and tidy results.

2 Related Work

Shapley-value feature attribution and its estimators. SHAP casts feature attribution as an additive Shapley-value problem [Lundberg and Lee, 2017]; exact computation is exponential in the number of features, so model-agnostic explainers estimate it by sampling coalitions (Kernel SHAP) or feature permutations. The variance of these estimators has been analysed and reduced by paired sampling and regression-based variants [Covert and Lee, 2021], by better permutation sampling [Mitchell et al., 2022], and by leverage-score sampling with error guarantees [Musco and Witter, 2025]. Sampling estimators of Shapley-type contributions go back to Štrumbelj and Kononenko [2014], and tree ensembles admit deterministic algorithms [Lundberg et al., 2020]. These works study estimator error as a function of the number of samples; we ask the deployment question that follows from them: given the budget a practitioner actually uses, *is one run enough to treat the attribution as the explanation of that prediction*, and for which instances and models.

What the target is. The quantity SHAP estimates depends on how the value function takes expectations over absent features and on the background data [Chen et al., 2020]; background-set size already affects explanation stability for neural models [Yuan et al., 2023]. We therefore define error

and agreement relative to a fixed model, value function and background set, and treat background resampling as a separate source of variation.

Disagreement and multiplicity. Explanations can disagree across methods [Krishna et al., 2024], and equally accurate models can disagree on individual predictions [Marx et al., 2020]. We use the second as a comparison standard (explainer-run noise is judged against between-refit variation instead of in isolation); between-method disagreement is not evaluated here.

Instability of sampling-based explainers. Other randomised explainers are known to vary between runs: for LIME, repeated runs can select different features, which motivated stability indices computed from repeated explanations [Visani et al., 2022], and robustness metrics for interpretability methods in general were proposed by Alvarez-Melis and Jaakkola [2018]. These works take repeatability as the criterion; for SHAP estimators we add comparison with an exact target and with other sources of variation.

Stability and agreement measures. Set-stability of selected features [Nogueira et al., 2018] and rank-biased overlap for top-weighted, incomplete rankings [Webber et al., 2010] give chance-aware ways to compare feature sets and rankings across runs; we adopt them alongside error and sign-agreement measures.

Positioning. The estimator-variance literature above gives error as a function of the number of samples for a given algorithm; the stability and multiplicity literature compares explanations across models or methods. The error-versus-samples curves of Figure 1 are in the spirit of Covert and Lee [2021], Mitchell et al. [2022] and Musco and Witter [2025]; what we add is their reading at the library's actual budgets for the question of top-feature recovery, together with the other notions below. The metrics themselves are not new; the combination is applied to one deployment question. We separate repeatability, accuracy against an exact target, resolvability of the ordering and noise relative to background and refit variation, evaluate them at default and neighbouring budgets of the most used implementation, and state the conditions under which the single-run practice holds. We read only the open literature cited here; other related work may exist that we did not consult.

3 Problem Setting and Reliability Protocol

3.1 Target, estimator and budget

Let $f : \mathbb{R}^d \to \mathbb{R}$ be a trained classifier's output (log-odds or probability; one choice is primary and the other is a sensitivity analysis), x an instance and D_b a background set of m training rows. We use the interventional value function $v_x(S) = \mathbb{E}_{Z\sim D_b}[f(x_S, Z_{\bar{S}})]$, i.e. the "true to the model" reading of Chen et al. [2020], and write $\boldsymbol{\phi}^\star(x) \in \mathbb{R}^d$ for its Shapley values. The players of the game are the *original attributes* of the data (not one-hot columns): for non-linear models a group value is not the sum of one-hot values, and one-hot columns under interventional expectation create impossible encodings. A stochastic explainer $\mathcal{E}_B$ with seed s returns $\hat{\boldsymbol{\phi}}_s(x)$, an estimate of $\boldsymbol{\phi}^\star(x)$ that uses a budget of B model evaluations (for Kernel SHAP, number of sampled coalitions $\times\, m$). The *single-run practice* we study is: draw one seed, explain, and report $\hat{\boldsymbol{\phi}}_s(x)$ as the attribution of the prediction. All randomness in $\hat{\boldsymbol{\phi}}_s$ for fixed (f, x, D_b) comes from s.

3.2 Four notions of reliability

"Reliable" is not one number. For an instance x and R independent runs $\hat{\boldsymbol{\phi}}_1, \ldots, \hat{\boldsymbol{\phi}}_R$ at the same budget, we separate:

N1 Repeatability. Do two independent runs agree? We report, over all run pairs, Spearman and Kendall correlation of the rankings by $|\hat{\phi}_j|$, top-k Jaccard overlap ($k \in \{1, 3, 5\}$), rank-biased overlap [Webber et al., 2010], sign agreement of the top-k features, and the normalised distance $\|\hat{\boldsymbol{\phi}}_a - \hat{\boldsymbol{\phi}}_b\|_2 / \|\boldsymbol{\phi}^\star\|_2$. Top-$k$ set stability over the R runs is summarised with the chance-corrected index of Nogueira et al. [2018]. A null reference is a random permutation of the features of the same attribution vector.

N2 Accuracy. Is one run close to the quantity it estimates? Relative ℓ_2 error $\|\hat{\boldsymbol{\phi}}_s - \boldsymbol{\phi}^\star\|_2 / \|\boldsymbol{\phi}^\star\|_2$, maximum absolute error, and top-k hit rate against $\boldsymbol{\phi}^\star$, which is exact where an exact algorithm for the same background exists and a converged high-budget reference with its own error bar otherwise.

N3 Resolvability. Is the noise small relative to the differences a reader would interpret? For adjacent features $(j), (j{+}1)$ in the reference ranking by $|\phi^\star|$, the gap-to-noise ratio is $\big(|\phi^\star_{(j)}| - |\phi^\star_{(j+1)}|\big) / \mathrm{SD}_s\big(|\hat{\phi}_{s,(j)}| - |\hat{\phi}_{s,(j+1)}|\big)$, together with the probability that a single run reproduces the reference order of the pair. The threshold $t{=}2$ is a convention; for Gaussian noise on the gap it corresponds to a pair-order error rate of about 2.3%. Gaps among small attributions are small in any attribution vector, so the fraction of resolved pairs is partly a property of the vector's spacing, and the order-reproduction probability is the more direct quantity.

N4 Relative size of the noise. Is explainer noise material next to other variation a deployed explanation already has? We compare the explainer-seed variance of $\hat{\phi}_j$ with (a) the variance from redrawing the background sample, (b) the variance across refits of the model with different seeds at comparable accuracy [Marx et al., 2020], as a decomposition into variance components. Disagreement between different explainers [Krishna et al., 2024] is a further reference scale that we do not evaluate.

Each verdict is a claim of the form "at budget B, for this dataset and model class, the single-run metric is μ (95% interval), and a fraction q of instances meets threshold τ"; thresholds are always swept rather than fixed.

Where a single run is trivially exact, and the shap default. Kernel SHAP enumerates all $2^d - 2$ coalitions once the number of samples reaches that number; for $d \leq 10$ the library default is therefore a deterministic computation and says nothing about sampling noise. The stochastic regime is d large relative to the budget, so our core datasets have $d \geq 16$ and low-dimensional data serve as a deterministic-regime control. Separately, `shap` 0.51.0 applies ℓ_1 feature selection (`l1_reg='num_features(10)'`) unconditionally in `KernelExplainer`; this is a deterministic bias with exact zeros that disturbs rank metrics at any budget when $d > 10$. We therefore report an ℓ_1-off arm as primary for N1–N3 and the library default arm separately at the default budget.

Decision rule. Specified in the study plan before the main runs (we cannot ship its time stamps in an anonymous submission, so read it as a convention) and applied per cell (dataset $\times$ model $\times$ budget): a single run is called *reliable for top-feature identification* if the lower 95% bound of the mean top-3 hit rate against the target is ≥ 0.9, the median relative ℓ_2 error is ≤ 0.1 and the lower bound of the mean top-3 sign agreement is ≥ 0.95; *unreliable* if the upper bound of the top-3 hit rate is < 0.7 or the upper bound of the top-1 Jaccard agreement between runs is < 0.5; *conditional* otherwise. These thresholds are conventions, and the rule is not powerful: when the top few features dominate, hit rates sit at their ceiling and most cells pass. We therefore report how the verdicts change under stricter thresholds (Appendix A.4); stricter notions (top-5 sets, the top-3 set in the target's order, shares of instances below Jaccard levels) are in Appendix A.5, and we treat the rule as a summary, not as a test.

3.3 Statistical treatment

The unit of inference is the instance. Repeats within an instance are not independent draws of instances, so all intervals are from a nested bootstrap that resamples instances and then runs within instance; pairwise statistics over the $R(R-1)/2$ pairs are used as point estimates only. Datasets are reported separately (there are few). Contrasts between models, explainers and budgets are paired over the same instances, reported with effect sizes and bootstrap intervals, with the planned contrasts listed in advance.

4 Experiments

4.1 Setup

Data. The stochastic core is German Credit (UCI, d=20 attributes, 1,000 rows, 700/300 stratified split) and ACSEmployment from folktables [Ding et al., 2021] (d=16; California, 2018 one-year person file, 40,000 rows subsampled with a fixed seed, 28,000/12,000 split). The deterministic-regime controls are ACSIncome (d=10) and Pima diabetes (d=8). Sources, versions and retrieval dates are in the reproduction notes.

Models. Logistic regression (LR), XGBoost (depth 4, 200 trees), a random forest (RF; 100 trees, depth ≤ 8) and an MLP (two hidden layers of 64 units), trained once with a fixed seed. Held-out AUC is 0.765–0.810 on German Credit and 0.874–0.905 on ACSEmployment (appendix table). Categorical attributes are ordinal-coded for trees; LR and MLP one-hot encode inside the pipeline so that masking an attribute masks its whole group. The explained value function is the log-odds for LR, XGBoost and MLP and the probability for RF, for which interventional TreeSHAP is exact; results are therefore not pooled across the RF and the other models. A sensitivity check with the other value function is in Appendix A.7.

Explainers, budget and targets. Kernel SHAP (`shap` 0.51.0 `KernelExplainer`) is the main explainer; `PermutationExplainer` is the second. The background is 50 training rows, fixed per dataset. The budget is counted in model evaluations, $B = \texttt{nsamples} \times 50$; the library default is $\texttt{nsamples} = 2d + 2048$ (e.g. B=104400 on German Credit, about 0.59 s per explanation on the shared CPU host used). We study $1/4$, 1 and 4 times the default for all cells and 16× on German Credit for 20 instances; the 16× ACSEmployment cell was cut for throughput. ℓ_1 selection is off in the primary arm; the library default is run separately at 1×. Exact targets use the same background and value function: closed form for LR, interventional TreeSHAP for XGBoost and RF, full enumeration for all ACS models, and enumeration on the first 20 instances for the German MLP (so its accuracy rows use 20 instances). Targets were cross-checked against enumeration (agreement to 2×10^{-16} for LR, 10^{-6} for XGBoost and 2×10^{-4} for RF).

Instances, runs and statistics. Fifty instances per dataset, shared by all models: 25 random test rows and 25 test rows nearest the decision boundary among a random pool of 300. Each cell has R=20 independent explainer seeds. Pairwise metrics use all 190 run pairs per instance. Intervals are 95% nested-bootstrap percentile intervals (1,000 draws: instances, then runs within instance); the instance is the unit of inference. Total compute was about 26 CPU-hours on a host oversubscribed during the runs. The null reference for agreement is a random permutation of the features of the same vector (chance level).

4.2 Single runs at the library default budget

With ℓ_1 off, one run is repeatable and accurate for top-feature identification. At 1× the mean top-3 hit rate against the exact target (per model, over 50 instances) is 0.977–1.000 on German Credit (lowest 95% lower bound 0.953) and 0.999–1.000 on ACSEmployment; the median relative ℓ_2 error is at most 0.0175 and 0.0038 respectively, and the mean pairwise Spearman correlation of $|\phi|$ rankings between runs is ≥ 0.990 and ≥ 0.993. LR is exact at machine precision at every budget (the model is additive), so its rows are a check on the metrics, not evidence about sampling noise; XGBoost and RF have an exact algorithm that one would use instead of sampling. The MLP is therefore the only model here for which sampling is the practical option: its top-3 hit rate at 1× is 0.977 [0.953, 0.995] on German Credit (20 instances, exact targets by enumeration) and 1.000 [0.999, 1.000] on ACSEmployment, with median relative error 0.0175 and 0.0035. All 4 German Credit and 4 ACSEmployment cells at 1× meet the reliability rule (Table 1); so do the $1/4$× cells (top-3 hit ≥ 0.954, relative error ≤ 0.0412 on German Credit), all 4× and 16× cells, and permutation SHAP at 1× and 4× (Table 1, Figure 1). Error falls with budget as expected (Figure 1). On the controls, the library default enumerates all coalitions, so repeatability is exact by construction and these cells are not evidence of reliability for larger d.

The shap default ℓ_1 setting is the exception. In `shap` 0.51.0 the default of `KernelExplainer.shap_values` is `l1_reg='num_features(10)'`, and the selection step

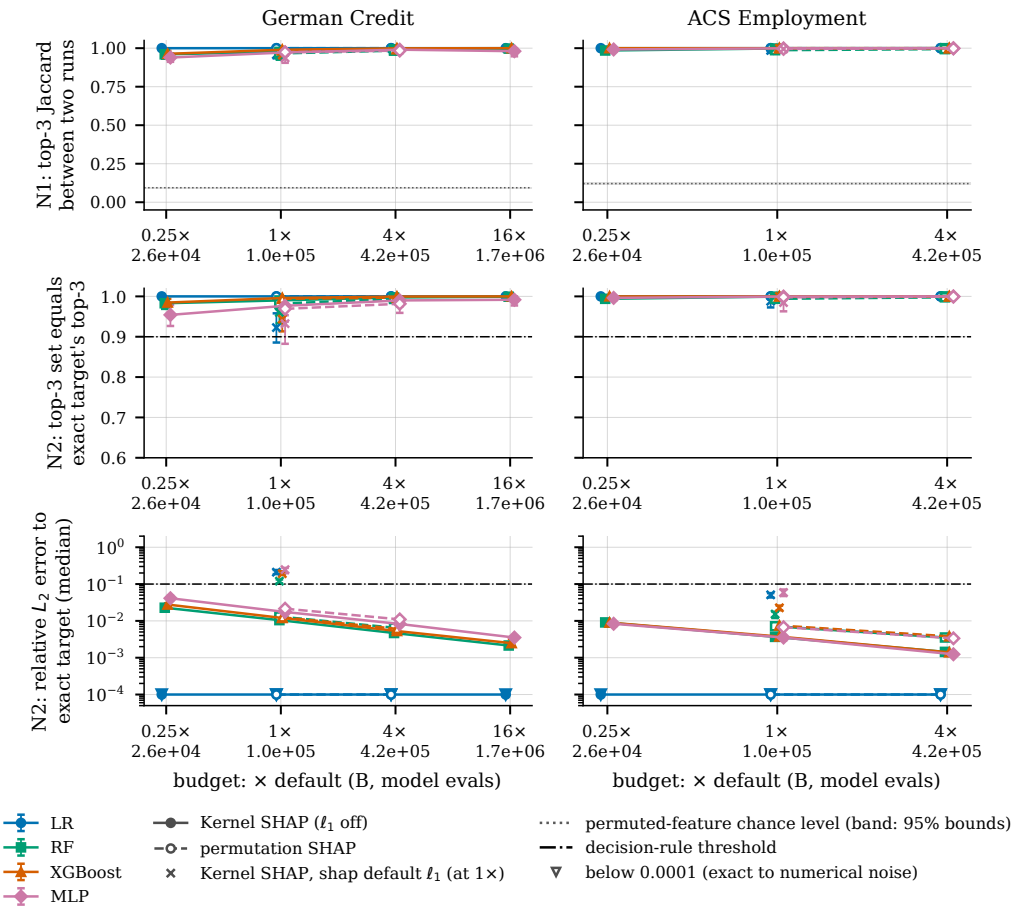


Figure 1: Single-run reliability of Kernel SHAP against budget on the two stochastic datasets (columns). The budget is the multiple of the library default `nsamples` $= 2d+2048$; $B =$ `nsamples`$\times$ 50 model evaluations is printed under each tick. Top row: N1, mean top-3 Jaccard between two runs. Middle: N2, fraction of explanations whose top-3 set equals the exact target's. Bottom: N2, median relative ℓ_2 error (log scale; triangles: below 10^{-4}, i.e. exact to numerical noise, as for the additive LR). Points are means over 50 instances (German MLP error rows: 20 instances) of per-instance means over runs; bars are 95% nested-bootstrap intervals. Dashed hollow: permutation SHAP at $1\times$ and $4\times$; crosses: the `shap` default ℓ_1 setting at $1\times$; dotted: permuted-feature chance level; dash-dotted: thresholds of the decision rule. $16\times$ exists for German Credit only.

runs whenever `l1_reg` is not `False`, `0` or `'auto'` (`shap/explainers/_kernel.py`), so it applies regardless of the sample count. On German Credit (d=20) the attributions of this arm have ten non-zero entries per instance (checked in the runs). This sparsification makes an error of this size expected, not a discovered failure of SHAP: it is a configuration pitfall of one implementation, and the arm was run at $1\times$ only. The median relative error against the exact target is 0.1199–0.2464, the top-3 hit rate 0.923–0.960, and all 4 cells are classed "conditional" (Table 1). Repeatability between runs stays high (Spearman 0.956–0.969) because the bias is shared by all runs, so a repeatability check alone would not detect it; this is why accuracy against an exact target is a separate notion. On ACSEmployment (d=16) the arm remains within the rule (relative error up to 0.0589).

The "four reliable" counts in Table 1 include LR (additive, exact) and tree models (exact target); only the MLP is a setting where sampling is the practical option. Under stricter thresholds the number of reliable MLP cells falls from 11 of 11 to 9 and 5 (Appendix A.4).

Averaging does not beat one larger run. At equal total evaluations, the mean of 4 runs at $B/4$ has a mean relative error that is higher than that of one run at B by 0.0007–0.0028 in absolute terms

Table 1: Verdicts of the pre-stated decision rule (Sec. 3): number of the four models classed reliable, conditional or unreliable in each cell. Cell `kernel_q`, `kernel_1`, `kernel_4`, `kernel_16`: Kernel SHAP at $1/4, 1, 4, 16\times$ the library budget with ℓ_1 off; `l1default`: the library default at $1\times$; `perm`: permutation SHAP. n=50 instances, R=20 runs (20 instances at $16\times$). ACS Income and Pima are deterministic-regime controls. Per-model rows and the cells not run are in Appendix A.3.

Dataset	Cell	reliable	conditional	unreliable
German Credit	kernel_q	4	0	0
German Credit	kernel_1	4	0	0
German Credit	kernel_4	4	0	0
German Credit	kernel_16	4	0	0
German Credit	kernel_1_l1default	0	4	0
German Credit	perm_1	4	0	0
German Credit	perm_4	4	0	0
ACS Income	kernel_q	4	0	0
ACS Income	kernel_1	4	0	0
ACS Income	kernel_4	4	0	0
ACS Income	kernel_1_l1default	4	0	0
ACS Employment	kernel_q	4	0	0
ACS Employment	kernel_1	4	0	0
ACS Employment	kernel_4	4	0	0
ACS Employment	kernel_1_l1default	4	0	0
ACS Employment	perm_1	4	0	0
ACS Employment	perm_4	4	0	0
Pima diabetes	kernel_q	4	0	0
Pima diabetes	kernel_1	4	0	0
Pima diabetes	kernel_4	4	0	0
Pima diabetes	kernel_1_l1default	4	0	0

(non-additive models; four $1/4\times$ runs against one $1\times$ run; four $1\times$ runs against one $4\times$ run: 0.0004–0.0010; bootstrap intervals exclude zero in every comparison); per-model effects and intervals in Table 9, while the top-3 hit rates differ by at most 0.001. The effect is small next to the error levels (0.004–0.02 at $1\times$), so we read it as "averaging did not help", not as a strong negative result. A plausible mechanism, not tested here, is a finite-sample bias of the Kernel SHAP regression that averaging does not remove; permutation SHAP was not compared (no $1/4\times$ permutation runs).

4.3 Where a single run is less reliable

Instances. The 50 instances are half random and half near the boundary, so the following are not estimates of a fraction of the test population. Averages hide a tail: on German Credit, 0.087 of the instance–model pairs of non-additive models have mean pairwise top-3 Jaccard below 0.9 at $1\times$ (the three models share instances, so these pairs are not independent), with a minimum of 0.737 (Appendix Figure 4); on ACSEmployment the minimum is 0.950. Hit rates against the exact target stay high even for these instances. By stratum, the mean top-3 hit rate is 0.997 (near boundary, 25 instances) against 0.984 (random, 25) on German Credit and 1.000 against 1.000 on ACSEmployment; with 25 instances per stratum we do not draw a conclusion about the boundary. Per-model strata and tail shares are in Tables 7 and 8, and stricter agreement notions (top-5, ordered top-3) in Table 6.

Resolvability of the ordering (N3). The top of the ranking is stable, but fine ordering is not. The fraction of adjacent pairs, in the exact target's ranking, whose gap exceeds twice the run-to-run standard deviation of that gap is 0.695–0.770 on German Credit and 0.852–0.950 on ACSEmployment at $1\times$ (non-additive models); for the first pair (ranks 1 vs 2) it is 0.950–1.000 and 1.000–1.000, but for pairs at ranks four and below on German Credit only 0.621–0.703. A single run reproduces the exact order of an adjacent pair with probability 0.935–0.947 (German Credit) and 0.967–0.990 (ACSEmployment) on average (Figure 2). The pairs that are not resolved are mostly at lower ranks, where gaps between small attributions are small in any vector, so the order-reproduction probability is the more direct number. Reading a single run's complete ranking therefore over-reports what it supports; the top-3 *set* is supported (Sec. 4.2), the order within it less so.

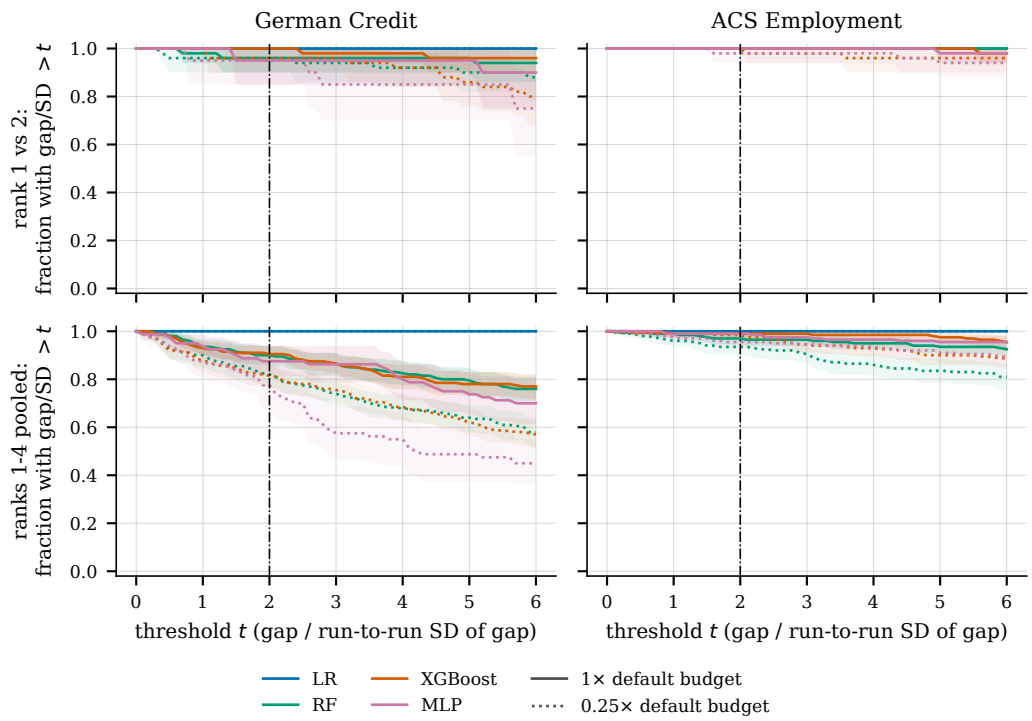


Figure 2: Resolvability of the ordering (N3). Fraction of adjacent-rank pairs of the exact target whose attribution gap exceeds t times the run-to-run standard deviation of that gap, as the threshold t varies (dash-dotted: $t = 2$). Top: ranks 1 versus 2; bottom: ranks 1–4 pooled. Solid: $1\times$; dotted: $1/4\times$ budget. Shaded: 95% instance-bootstrap intervals (German MLP: 20 instances, hence wider). Controls are omitted (every pair is resolvable).

Noise relative to other variation (N4; descriptive, at a background of 50 rows and $1\times$). Redrawing the 50-row background changes the attribution vector by 15.2–19.1% of its norm on German Credit and 20.3–22.0% on ACSEmployment (standard deviation over 10 redraws, 3 explainer seeds per redraw), against 1.2–2.0% and 0.6–0.7% from the explainer seed (20 instances; non-additive models). As variances, the background factor is 81–152 and 843–1102 times larger, and refitting the model with 10 other seeds gives ratios of 133–391 and 134–4465 (Figure 3). A background redraw changes the *target* Shapley vector rather than the estimate of a fixed one, so this ratio shows that the explainer seed is a minor part of the variation a deployed explanation carries; it does not show that the explainer is accurate for some "correct" background; refit ratios for LR are undefined (identical models) and the probe has 20 instances. Ratios with intervals per model are in Table 11.

5 Discussion and Limitations

Answer to the question. The verdict is *conditional, and mostly favourable* for the sampling noise itself. In the regime where Kernel SHAP and permutation SHAP are genuinely stochastic ($d = 16$ and $d = 20$) and at budgets from a quarter of the library default upward, one run identifies the top-3 features of a prediction, its signs and its overall magnitude profile reliably by the pre-stated rule. The reliability does not extend to (i) the library's default ℓ_1 setting for $d > 10$, which introduces a bias that repetition cannot reveal (this is a pitfall of one implementation, shown at one budget), (ii) the ordering of features whose attributions are close (on German Credit, only 0.621–0.703 of the pairs at ranks four and below are resolved at twice the noise), and (iii) a minority of instances with markedly lower agreement between runs. For low-dimensional data the question is moot because the default is exact. A practitioner who cannot repeat the explainer should (a) switch off ℓ_1 selection or check it against an exact or high-budget reference on a sample of instances, (b) read the few leading features rather than a full ranking, and (c) be aware that at a 50-row background the choice of

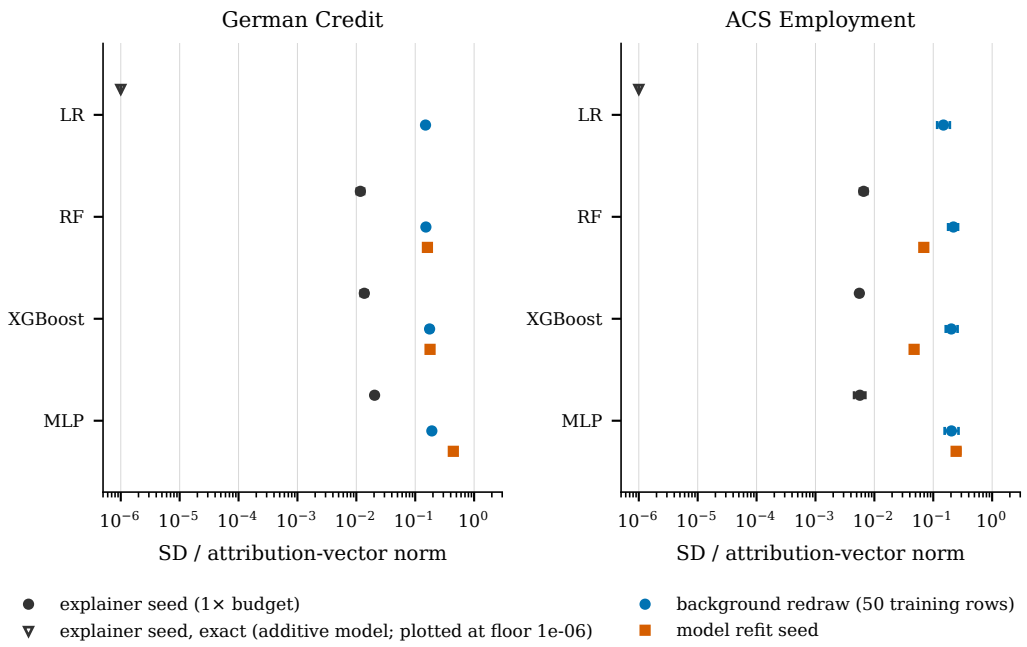


Figure 3: Noise from three sources (N4): standard deviation of the attribution vector, as a fraction of its norm, across explainer seeds at $1\times$ budget, across 10 redraws of the 50-row background, and across 10 refits of the model (20 instances; the last two with 3 explainer seeds each). Bars: 95% bootstrap intervals over instances. LR has exact Kernel SHAP, so its explainer noise is drawn at the floor (10^{-6}); its refit component is zero by construction and is not drawn. A background redraw changes the target vector itself.

background changed the attribution vector much more than the explainer seed did (the ratio depends on the background size and the budget, neither of which we varied). If model evaluation is fast, $16\times$ the default costs about 5.27 s per explanation on German Credit on our shared host and lowers the already small error further (Figure 1).

Limitations. The claims are limited to the following. *Few stochastic datasets:* two ($d = 20$ with 1,000 rows; $d = 16$ with a 40,000-row subsample of one state-year), so dataset-level uncertainty is not estimated; larger d (e.g. hundreds of features) is not covered and plausibly changes the answer. *Fixed design choices:* one background size (50 rows), one model fit per model class (refit variation is a 10-seed probe on 20 instances), small non-tuned model configurations, an interventional value function, one library (`shap` 0.51.0) and its implementation of Kernel SHAP and permutation SHAP; other estimators [Covert and Lee, 2021, Musco and Witter, 2025] are not evaluated. *Targets:* accuracy is against exact targets for the same background; it is not accuracy with respect to any ground truth of "importance", and a background redraw changes the target. The German MLP accuracy rows use 20 instances, and the $16\times$ cell uses 20 instances on German Credit only. *Instances and thresholds:* 50 instances per dataset, half chosen near the decision boundary; the decision-rule thresholds are conventions fixed before the runs (they are not tuned), and the fraction-over-threshold results should be read over their sweep. *Compute:* the budgets are CPU-scale; wall-clock numbers come from a shared, oversubscribed host and are not benchmarks. *Not run:* the background-size axis, comparisons with other explainers and higher-dimensional datasets (Give Me Some Credit, Arrhythmia) were cut for budget reasons and are not reported as done.

Broader impact. Attributions support high-stakes decisions. Our results suggest that a single run is rarely the weakest link at realistic budgets, which should not be read as a reason to trust attributions in general: the conclusions concern estimator noise for a fixed model, background and value function, not whether the explanation is faithful, causal or fair.

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A Appendix

A.1 Models and datasets

Table 2 lists the explained models. Training uses a stratified 70/30 split with a fixed seed; the models are not tuned. XGBoost: depth 4, 200 trees, learning rate 0.1, subsample 0.8; random forest: 100 trees, depth at most 8, minimum leaf size 3; MLP: two hidden layers of 64 units, L_2 penalty 10^{-2}; LR: $C{=}1$. The 50-row background is drawn from the training rows with a fixed seed. The 50 instances per dataset are 25 random test rows and the 25 of a random pool of 300 test rows nearest to the decision boundary.

Table 2: Explained models, the value function explained, and held-out performance (one fit per model, seed 0).

Dataset	Model	d	n_{train} / n_{test}	explained output	test AUC	test acc.
German Credit	LR	20	700 / 300	log-odds	0.805	0.760
German Credit	RF	20	700 / 300	probability	0.810	0.757
German Credit	XGBoost	20	700 / 300	log-odds	0.785	0.760
German Credit	MLP	20	700 / 300	log-odds	0.765	0.760
ACS Income	LR	10	28000 / 12000	log-odds	0.893	0.814
ACS Income	RF	10	28000 / 12000	probability	0.889	0.810
ACS Income	XGBoost	10	28000 / 12000	log-odds	0.900	0.819
ACS Income	MLP	10	28000 / 12000	log-odds	0.899	0.817
ACS Employment	LR	16	28000 / 12000	log-odds	0.874	0.797
ACS Employment	RF	16	28000 / 12000	probability	0.900	0.814
ACS Employment	XGBoost	16	28000 / 12000	log-odds	0.905	0.824
ACS Employment	MLP	16	28000 / 12000	log-odds	0.900	0.815
Pima diabetes	LR	8	537 / 231	log-odds	0.871	0.775
Pima diabetes	RF	8	537 / 231	probability	0.848	0.771
Pima diabetes	XGBoost	8	537 / 231	log-odds	0.821	0.753
Pima diabetes	MLP	8	537 / 231	log-odds	0.830	0.749

A.2 Controls, per-instance distributions

Figure 4 shows, for each explained instance on the stochastic datasets, the mean top-3 Jaccard between two runs at $1/4\times$ and $1\times$. Figure 5 repeats the budget curves on the two deterministic-regime controls, where the library default enumerates all coalitions and the budget curves are flat by construction.

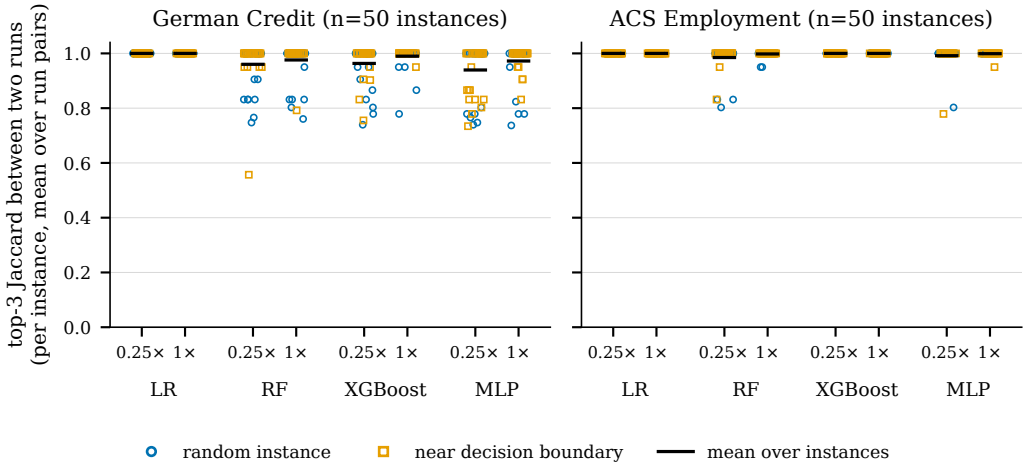


Figure 4: Per-instance mean pairwise top-3 Jaccard between runs (190 run pairs per instance) at $1/4\times$ and $1\times$ budget, for every explained instance, coloured by stratum (random or near boundary); black bar: mean. A minority of instances lie well below the mean in both strata.

A.3 Per-model verdicts and cells not run

Table 3 gives, for every cell, the metrics entering the decision rule; Table 4 lists the cells not run and why. Not run at all: the background-size axis, the Give Me Some Credit and Arrhythmia datasets, and comparisons with other explainers; they were cut for compute budget before the runs started and nothing was run and discarded, except that partially finished $16\times$ ACS Employment cell files were deleted.

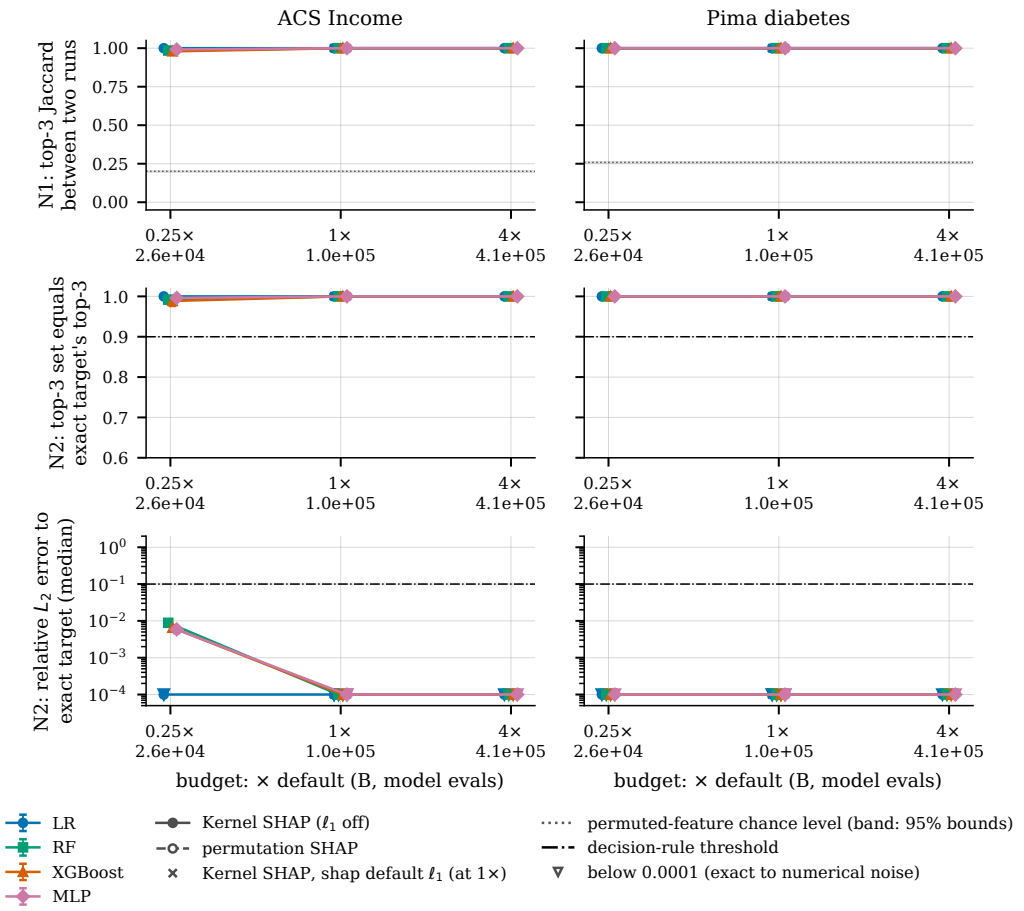


Figure 5: As Figure 1 for the controls ACS Income (d=10) and Pima diabetes (d=8), where coalition enumeration is used once the budget allows it.

Dataset	Model	Cell	B	n/R	top-3 hit [95% CI]	median rel. L_2	sign lo	top-1 Jacc.	verdict
German Credit	LR	kernel_q	26100	50/20	1.00 [1.00, 1.00]	1.1e-15	1.00	1.00	reliable
German Credit	LR	kernel_1	104400	50/20	1.00 [1.00, 1.00]	2.5e-15	1.00	1.00	reliable
German Credit	LR	kernel_4	417600	50/20	1.00 [1.00, 1.00]	3.4e-15	1.00	1.00	reliable
German Credit	LR	kernel_16	1670400	20/20	1.00 [1.00, 1.00]	2.9e-15	1.00	1.00	reliable
German Credit	LR	kernel_1_l1default	104400	50/20	0.92 [0.89, 0.96]	0.21	1.00	0.97	conditional
German Credit	LR	perm_1	104400	50/20	1.00 [1.00, 1.00]	9.2e-16	1.00	1.00	reliable
German Credit	LR	perm_4	417600	50/20	1.00 [1.00, 1.00]	4.8e-15	1.00	1.00	reliable
German Credit	RF	kernel_q	26100	50/20	0.98 [0.97, 0.99]	0.023	1.00	0.98	reliable
German Credit	RF	kernel_1	104400	50/20	0.99 [0.98, 1.00]	0.01	1.00	0.99	reliable
German Credit	RF	kernel_4	417600	50/20	1.00 [0.99, 1.00]	0.0047	1.00	1.00	reliable
German Credit	RF	kernel_16	1670400	20/20	1.00 [1.00, 1.00]	0.0021	1.00	1.00	reliable
German Credit	RF	kernel_1_l1default	104400	50/20	0.96 [0.93, 0.98]	0.12	1.00	0.99	conditional
German Credit	RF	perm_1	104400	50/20	0.98 [0.97, 0.99]	0.013	1.00	0.98	reliable
German Credit	RF	perm_4	417600	50/20	0.99 [0.99, 1.00]	0.0065	1.00	0.99	reliable
German Credit	XGBoost	kernel_q	26100	50/20	0.98 [0.97, 0.99]	0.028	1.00	0.99	reliable
German Credit	XGBoost	kernel_1	104400	50/20	1.00 [0.99, 1.00]	0.012	1.00	1.00	reliable
German Credit	XGBoost	kernel_4	417600	50/20	1.00 [1.00, 1.00]	0.0053	1.00	1.00	reliable
German Credit	XGBoost	kernel_16	1670400	20/20	1.00 [1.00, 1.00]	0.0025	1.00	1.00	reliable
German Credit	XGBoost	kernel_1_l1default	104400	50/20	0.94 [0.91, 0.97]	0.19	0.98	0.99	conditional
German Credit	XGBoost	perm_1	104400	50/20	0.99 [0.99, 1.00]	0.012	1.00	1.00	reliable
German Credit	XGBoost	perm_4	417600	50/20	1.00 [0.99, 1.00]	0.0061	1.00	1.00	reliable
German Credit	MLP	kernel_q	26100	50/20	0.95 [0.93, 0.98]	0.041	1.00	0.96	reliable
German Credit	MLP	kernel_1	104400	50/20	0.98 [0.95, 0.99]	0.017	1.00	0.99	reliable
German Credit	MLP	kernel_4	417600	50/20	0.99 [0.97, 1.00]	0.0082	1.00	1.00	reliable
German Credit	MLP	kernel_16	1670400	20/20	0.99 [0.98, 1.00]	0.0035	1.00	1.00	reliable
German Credit	MLP	kernel_1_l1default	104400	50/20	0.93 [0.88, 0.97]	0.25	0.95	0.98	conditional
German Credit	MLP	perm_1	104400	50/20	0.97 [0.94, 0.99]	0.021	1.00	0.97	reliable
German Credit	MLP	perm_4	417600	50/20	0.98 [0.96, 1.00]	0.011	1.00	1.00	reliable

Dataset	Model	Cell	B	n/R	top-3 hit [95% CI]	median rel. L_2	sign lo	top-1 Jacc.	verdict
ACS Income	LR	kernel_q	25850	50/20	1.00 [1.00, 1.00]	9.8e-16	1.00	1.00	reliable
ACS Income	LR	kernel_1	103400	50/20	1.00 [1.00, 1.00]	2.3e-15	1.00	1.00	reliable
ACS Income	LR	kernel_4	413600	50/20	1.00 [1.00, 1.00]	2.3e-15	1.00	1.00	reliable
ACS Income	LR	kernel_1_l1default	103400	50/20	1.00 [1.00, 1.00]	2.1e-15	1.00	1.00	reliable
ACS Income	RF	kernel_q	25850	50/20	0.99 [0.98, 1.00]	0.0089	1.00	1.00	reliable
ACS Income	RF	kernel_1	103400	50/20	1.00 [1.00, 1.00]	2.6e-15	1.00	1.00	reliable
ACS Income	RF	kernel_4	413600	50/20	1.00 [1.00, 1.00]	2.6e-15	1.00	1.00	reliable
ACS Income	RF	kernel_1_l1default	103400	50/20	1.00 [1.00, 1.00]	2.3e-15	1.00	1.00	reliable
ACS Income	XGBoost	kernel_q	25850	50/20	0.99 [0.98, 1.00]	0.0065	1.00	0.99	reliable
ACS Income	XGBoost	kernel_1	103400	50/20	1.00 [1.00, 1.00]	4.9e-08	1.00	1.00	reliable
ACS Income	XGBoost	kernel_4	413600	50/20	1.00 [1.00, 1.00]	4.9e-08	1.00	1.00	reliable
ACS Income	XGBoost	kernel_1_l1default	103400	50/20	1.00 [1.00, 1.00]	4.9e-08	1.00	1.00	reliable
ACS Income	MLP	kernel_q	25850	50/20	1.00 [0.99, 1.00]	0.0059	1.00	1.00	reliable
ACS Income	MLP	kernel_1	103400	50/20	1.00 [1.00, 1.00]	2.4e-15	1.00	1.00	reliable
ACS Income	MLP	kernel_4	413600	50/20	1.00 [1.00, 1.00]	2.4e-15	1.00	1.00	reliable
ACS Income	MLP	kernel_1_l1default	103400	50/20	1.00 [1.00, 1.00]	2.2e-15	1.00	1.00	reliable
ACS Employment	LR	kernel_q	26000	50/20	1.00 [1.00, 1.00]	9.2e-16	1.00	1.00	reliable
ACS Employment	LR	kernel_1	104000	50/20	1.00 [1.00, 1.00]	3.3e-15	1.00	1.00	reliable
ACS Employment	LR	kernel_4	416000	50/20	1.00 [1.00, 1.00]	2.4e-15	1.00	1.00	reliable
ACS Employment	LR	kernel_1_l1default	104000	50/20	0.99 [0.97, 1.00]	0.051	1.00	0.99	reliable
ACS Employment	LR	perm_1	104000	50/20	1.00 [1.00, 1.00]	1.6e-15	1.00	1.00	reliable
ACS Employment	LR	perm_4	416000	50/20	1.00 [1.00, 1.00]	5.4e-15	1.00	1.00	reliable
ACS Employment	RF	kernel_q	26000	50/20	0.99 [0.99, 1.00]	0.0091	1.00	1.00	reliable
ACS Employment	RF	kernel_1	104000	50/20	1.00 [1.00, 1.00]	0.0037	1.00	1.00	reliable
ACS Employment	RF	kernel_4	416000	50/20	1.00 [1.00, 1.00]	0.0014	1.00	1.00	reliable
ACS Employment	RF	kernel_1_l1default	104000	50/20	1.00 [1.00, 1.00]	0.015	1.00	1.00	reliable
ACS Employment	RF	perm_1	104000	50/20	0.99 [0.99, 1.00]	0.0072	1.00	1.00	reliable
ACS Employment	RF	perm_4	416000	50/20	1.00 [0.99, 1.00]	0.0035	1.00	1.00	reliable
ACS Employment	XGBoost	kernel_q	26000	50/20	1.00 [1.00, 1.00]	0.009	1.00	1.00	reliable
ACS Employment	XGBoost	kernel_1	104000	50/20	1.00 [1.00, 1.00]	0.0038	1.00	1.00	reliable
ACS Employment	XGBoost	kernel_4	416000	50/20	1.00 [1.00, 1.00]	0.0014	1.00	1.00	reliable
ACS Employment	XGBoost	kernel_1_l1default	104000	50/20	1.00 [1.00, 1.00]	0.023	1.00	1.00	reliable
ACS Employment	XGBoost	perm_1	104000	50/20	1.00 [1.00, 1.00]	0.0075	1.00	1.00	reliable
ACS Employment	XGBoost	perm_4	416000	50/20	1.00 [1.00, 1.00]	0.0038	1.00	1.00	reliable
ACS Employment	MLP	kernel_q	26000	50/20	1.00 [0.99, 1.00]	0.0084	1.00	0.99	reliable
ACS Employment	MLP	kernel_1	104000	50/20	1.00 [1.00, 1.00]	0.0035	1.00	1.00	reliable
ACS Employment	MLP	kernel_4	416000	50/20	1.00 [1.00, 1.00]	0.0012	1.00	1.00	reliable
ACS Employment	MLP	kernel_1_l1default	104000	50/20	0.98 [0.96, 1.00]	0.059	1.00	1.00	reliable
ACS Employment	MLP	perm_1	104000	50/20	1.00 [0.99, 1.00]	0.0067	1.00	0.99	reliable
ACS Employment	MLP	perm_4	416000	50/20	1.00 [1.00, 1.00]	0.0034	1.00	1.00	reliable
Pima diabetes	LR	kernel_q	25800	50/20	1.00 [1.00, 1.00]	6.1e-16	1.00	1.00	reliable
Pima diabetes	LR	kernel_1	103200	50/20	1.00 [1.00, 1.00]	6.1e-16	1.00	1.00	reliable
Pima diabetes	LR	kernel_4	412800	50/20	1.00 [1.00, 1.00]	6.1e-16	1.00	1.00	reliable
Pima diabetes	LR	kernel_1_l1default	103200	50/20	1.00 [1.00, 1.00]	6.2e-16	1.00	1.00	reliable
Pima diabetes	RF	kernel_q	25800	50/20	1.00 [1.00, 1.00]	6.2e-16	1.00	1.00	reliable
Pima diabetes	RF	kernel_1	103200	50/20	1.00 [1.00, 1.00]	6.2e-16	1.00	1.00	reliable
Pima diabetes	RF	kernel_4	412800	50/20	1.00 [1.00, 1.00]	6.2e-16	1.00	1.00	reliable
Pima diabetes	RF	kernel_1_l1default	103200	50/20	1.00 [1.00, 1.00]	7.2e-16	1.00	1.00	reliable
Pima diabetes	XGBoost	kernel_q	25800	50/20	1.00 [1.00, 1.00]	3.3e-08	1.00	1.00	reliable
Pima diabetes	XGBoost	kernel_1	103200	50/20	1.00 [1.00, 1.00]	3.3e-08	1.00	1.00	reliable
Pima diabetes	XGBoost	kernel_4	412800	50/20	1.00 [1.00, 1.00]	3.3e-08	1.00	1.00	reliable
Pima diabetes	XGBoost	kernel_1_l1default	103200	50/20	1.00 [1.00, 1.00]	3.3e-08	1.00	1.00	reliable
Pima diabetes	MLP	kernel_q	25800	50/20	1.00 [1.00, 1.00]	4.5e-16	1.00	1.00	reliable
Pima diabetes	MLP	kernel_1	103200	50/20	1.00 [1.00, 1.00]	4.5e-16	1.00	1.00	reliable
Pima diabetes	MLP	kernel_4	412800	50/20	1.00 [1.00, 1.00]	4.5e-16	1.00	1.00	reliable
Pima diabetes	MLP	kernel_1_l1default	103200	50/20	1.00 [1.00, 1.00]	5.5e-16	1.00	1.00	reliable

Table 3: Per cell: budget B (model evaluations), instances/runs, top-3 hit rate against the exact target with its 95% interval, median relative L_2 error, lower bound of top-3 sign agreement, top-1 Jaccard between runs, and the verdict.

Dataset	Model	Cell	Reason
ACS Income	LR	kernel_16	not planned (deterministic-regime control: Tier 1 only)
ACS Income	LR	perm_1	not planned (deterministic-regime control: Tier 1 only)
ACS Income	LR	perm_4	not planned (deterministic-regime control: Tier 1 only)
ACS Income	RF	kernel_16	not planned (deterministic-regime control: Tier 1 only)
ACS Income	RF	perm_1	not planned (deterministic-regime control: Tier 1 only)
ACS Income	RF	perm_4	not planned (deterministic-regime control: Tier 1 only)
ACS Income	XGBoost	kernel_16	not planned (deterministic-regime control: Tier 1 only)
ACS Income	XGBoost	perm_1	not planned (deterministic-regime control: Tier 1 only)
ACS Income	XGBoost	perm_4	not planned (deterministic-regime control: Tier 1 only)
ACS Income	MLP	kernel_16	not planned (deterministic-regime control: Tier 1 only)
ACS Income	MLP	perm_1	not planned (deterministic-regime control: Tier 1 only)
ACS Income	MLP	perm_4	not planned (deterministic-regime control: Tier 1 only)

Dataset	Model	Cell	Reason
ACS Employment	LR	kernel_16	cut for throughput (PLAN §0a.6 cut order); 16x kept on German only
ACS Employment	RF	kernel_16	cut for throughput (PLAN §0a.6 cut order); 16x kept on German only
ACS Employment	XGBoost	kernel_16	cut for throughput (PLAN §0a.6 cut order); 16x kept on German only
ACS Employment	MLP	kernel_16	cut for throughput (PLAN §0a.6 cut order); 16x kept on German only
Pima diabetes	LR	kernel_16	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	LR	perm_1	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	LR	perm_4	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	RF	kernel_16	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	RF	perm_1	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	RF	perm_4	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	XGBoost	kernel_16	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	XGBoost	perm_1	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	XGBoost	perm_4	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	MLP	kernel_16	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	MLP	perm_1	not planned (deterministic-regime control: Tier 1 only)
Pima diabetes	MLP	perm_4	not planned (deterministic-regime control: Tier 1 only)

Table 4: Cells not run, with reasons.

A.4 Sensitivity of the verdicts to the thresholds

Table 5 recounts the cells (reliable / conditional / unreliable) when the thresholds of the rule are varied one at a time and together, on German Credit and ACS Employment, over all arms including the ℓ_1-default arm. The counts are computed from the bounds already in `verdicts.csv`; the pre-stated row reproduces the stored verdicts, and the cells are not independent (the same instances serve every budget and arm). The additive LR is exact in every cell and the tree models are explained with an exact target, so only the MLP column is evidence about a setting where sampling is the practical option. With ℓ_1 off, the number of reliable MLP cells falls from 11 of 11 under the pre-stated rule to 9 and 5 under the two strict sets; on German Credit all six MLP cells fail the strictest set (lowest hit-rate lower bound 0.927, largest median relative error 0.041). The verdict "reliable" is therefore threshold-sensitive for the one informative model.

Table 5: Cells classed reliable / conditional / unreliable under alternative threshold sets (h, e, s) (top-3 hit lower bound, median relative L_2, sign-agreement lower bound; the criteria for "unreliable" are unchanged), for all models, without LR, and for the MLP alone. The number of cells is in the header.

Thresholds (h, e, s)	all models (52)	without LR (39)	MLP only (13)
pre-stated (0.90, 0.10, 0.95)	48 / 4 / 0	36 / 3 / 0	12 / 1 / 0
hit 0.95	46 / 6 / 0	34 / 5 / 0	10 / 3 / 0
hit 0.99	32 / 20 / 0	21 / 18 / 0	5 / 8 / 0
L2 0.05	46 / 6 / 0	35 / 4 / 0	11 / 2 / 0
L2 0.02	41 / 11 / 0	30 / 9 / 0	9 / 4 / 0
sign 0.99	48 / 4 / 0	36 / 3 / 0	12 / 1 / 0
strict (0.95, 0.05, 0.99)	44 / 8 / 0	33 / 6 / 0	9 / 4 / 0
strictest (0.99, 0.02, 0.99)	31 / 21 / 0	20 / 19 / 0	5 / 8 / 0

A.5 Stricter notions, strata and tails

Table 6 reports, at $1\times$ with ℓ_1 off, stricter agreement notions than the top-3 set: top-1, top-3 and top-5 hit, the top-3 set in the target's order, and the sign of all features. Ordering is the weakest notion: the ordered top-3 hit is 0.927 for the German MLP (20 instances) against 0.977 for the unordered set. Tables 7 and 8 split the 50 instances into the two sampling strata (25 random, 25 near the boundary) and give the share of instances whose per-instance mean top-3 Jaccard falls below 0.99 to 0.80. The explained set is a balanced mix by design, so neither pooled value is a population fraction, and with 25 instances per stratum the intervals are wide; we draw no conclusion about which stratum is harder. Not computed: the per-(instance $\times$ run) top-3 hit fraction and the top-10 hit.

Dataset	Model	Stratum	n	Spearman $\lvert\phi\rvert$	top-3 Jaccard	top-3 hit	rel. L_2
German	LR	random	25	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	2.5e-15 [2.2e-15, 2.8e-15]
German	LR	near boundary	25	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	2.5e-15 [2.4e-15, 2.7e-15]

Dataset	Model	Stratum	n	Spearman $\lvert\phi\rvert$	top-3 Jaccard	top-3 hit	rel. L_2
German	RF	random	25	0.992 [0.991, 0.994]	0.960 [0.928, 0.989]	0.983 [0.969, 0.994]	0.013 [0.011, 0.015]
German	RF	near boundary	25	0.992 [0.990, 0.993]	0.992 [0.975, 1.000]	0.997 [0.990, 1.000]	0.011 [0.0096, 0.012]
German	XGBoost	random	25	0.992 [0.989, 0.994]	0.982 [0.957, 0.998]	0.993 [0.982, 0.999]	0.014 [0.013, 0.017]
German	XGBoost	near boundary	25	0.994 [0.993, 0.995]	0.998 [0.994, 1.000]	0.999 [0.998, 1.000]	0.012 [0.011, 0.014]
German	MLP	random	25	0.992 [0.990, 0.993]	0.963 [0.926, 0.991]	0.962 [0.925, 0.990]	0.018 [0.017, 0.02]
German	MLP	near boundary	25	0.989 [0.987, 0.992]	0.982 [0.963, 0.996]	0.992 [0.977, 1.000]	0.021 [0.017, 0.025]
ACS Emp.	LR	random	25	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	3.5e-15 [3.3e-15, 3.7e-15]
ACS Emp.	LR	near boundary	25	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	3.3e-15 [3.2e-15, 3.5e-15]
ACS Emp.	RF	random	25	0.992 [0.989, 0.994]	0.996 [0.990, 1.000]	0.999 [0.997, 1.000]	0.0034 [0.0029, 0.0038]
ACS Emp.	RF	near boundary	25	0.994 [0.992, 0.996]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.0046 [0.004, 0.0052]
ACS Emp.	XGBoost	random	25	0.994 [0.992, 0.996]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.0036 [0.0033, 0.004]
ACS Emp.	XGBoost	near boundary	25	0.995 [0.994, 0.996]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.0047 [0.0041, 0.0054]
ACS Emp.	MLP	random	25	0.998 [0.998, 0.999]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.0033 [0.0028, 0.0039]
ACS Emp.	MLP	near boundary	25	0.999 [0.998, 0.999]	0.998 [0.994, 1.000]	0.999 [0.998, 1.000]	0.0037 [0.0033, 0.0041]

Table 7: Main $1\times$ metrics per instance stratum (random versus near the decision boundary), instance-bootstrap 95% intervals within each stratum (1000 draws); German MLP error rows rest on fewer instances.

Dataset	Model	Stratum	n	< 0.99	< 0.95	< 0.90	< 0.80
German	LR	random	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
German	LR	near boundary	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
German	RF	random	25	24 [8, 40]	20 [8, 36]	20 [4, 36]	4 [0, 12]
German	RF	near boundary	25	4 [0, 12]	4 [0, 12]	4 [0, 12]	4 [0, 12]
German	XGBoost	random	25	16 [4, 32]	8 [0, 20]	8 [0, 20]	4 [0, 12]
German	XGBoost	near boundary	25	4 [0, 12]	0 [0, 0]	0 [0, 0]	0 [0, 0]
German	MLP	random	25	20 [4, 36]	16 [4, 32]	16 [4, 32]	12 [0, 24]
German	MLP	near boundary	25	20 [8, 36]	12 [0, 24]	4 [0, 12]	0 [0, 0]
ACS Emp.	LR	random	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	LR	near boundary	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	RF	random	25	8 [0, 20]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	RF	near boundary	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	XGBoost	random	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	XGBoost	near boundary	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	MLP	random	25	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
ACS Emp.	MLP	near boundary	25	4 [0, 12]	0 [0, 0]	0 [0, 0]	0 [0, 0]

Table 8: Percentage of instances whose per-instance mean top-3 Jaccard between runs is below 0.99, 0.95, 0.90 and 0.80, per stratum, with instance-bootstrap intervals.

A.6 Budget contrasts, equal-budget comparison and N4 ratios

Tables 9–11 give, for every cell, the paired contrasts between budgets, the equal-budget averaging comparison and the N4 variance ratios with their intervals. Intervals are nested-bootstrap percentile intervals (instances, then runs); Holm p-values correct over the contrasts of one family. Rows of exact or additive models (LR, and tree models on the exact enumeration cells) are included for completeness; their contrasts are at floating-point level and carry no evidence about sampling noise.

Dataset	Model	Averaged vs single	n	Δ top-3 hit [95% CI]	single-run rel. L_2	Δ rel. L_2 [95% CI]
German	LR	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	2.5e-15	-1.9e-15 [-2e-15, -1.8e-15]
German	LR	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	3.4e-15	-2e-15 [-2.1e-15, -1.9e-15]
German	RF	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.001 [-0.010, 0.008]	0.012	0.0014 [0.001, 0.0018]
German	RF	$4\times1\times$ vs $4\times$	50	0.000 [-0.004, 0.004]	0.0052	0.00055 [0.00038, 0.00075]
German	XGBoost	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.001 [-0.005, 0.002]	0.013	0.0016 [0.0012, 0.0021]
German	XGBoost	$4\times1\times$ vs $4\times$	50	-0.001 [-0.003, 0.000]	0.0059	0.00084 [0.00068, 0.001]
German	MLP	$4\times\frac{1}{4}\times$ vs $1\times$	20	0.000 [-0.016, 0.017]	0.02	0.0028 [0.002, 0.0037]
German	MLP	$4\times1\times$ vs $4\times$	20	0.000 [-0.008, 0.007]	0.0086	0.001 [0.00054, 0.0016]
ACS Inc.	LR	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	2.5e-15	-1.9e-15 [-2.1e-15, -1.7e-15]
ACS Inc.	LR	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	2.5e-15	-5.5e-32 [-1.6e-31, 4.7e-32]
ACS Inc.	RF	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.004 [-0.011, 0.000]	2.7e-15	0.0048 [0.0043, 0.0055]
ACS Inc.	RF	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	2.7e-15	-5.9e-32 [-1.9e-31, 7.1e-32]
ACS Inc.	XGBoost	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.007 [-0.017, 0.000]	5.4e-08	0.0035 [0.0031, 0.0039]
ACS Inc.	XGBoost	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	5.4e-08	-1.3e-24 [-3.9e-24, 1.8e-24]
ACS Inc.	MLP	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.001 [-0.004, 0.000]	2.5e-15	0.0032 [0.0028, 0.0037]
ACS Inc.	MLP	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	2.5e-15	0 [-1.1e-31, 1.1e-31]
ACS Emp.	LR	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	3.4e-15	-2.9e-15 [-3e-15, -2.8e-15]
ACS Emp.	LR	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	2.5e-15	-7e-16 [-8.5e-16, -5.6e-16]
ACS Emp.	RF	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.001 [0.000, 0.002]	0.004	0.00079 [0.00064, 0.00095]
ACS Emp.	RF	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	0.0015	0.00051 [0.00044, 0.00059]

Dataset	Model	Averaged vs single	n	Δ top-3 hit [95% CI]	single-run rel. L_2	Δ rel. L_2 [95% CI]
ACS Emp.	XGBoost	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0.0042	0.00074 [0.00062, 0.00087]
ACS Emp.	XGBoost	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	0.0016	0.00052 [0.00045, 0.00059]
ACS Emp.	MLP	$4\times\frac{1}{4}\times$ vs $1\times$	50	-0.001 [-0.003, 0.000]	0.0035	0.00098 [0.00084, 0.0012]
ACS Emp.	MLP	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	0.0013	0.00044 [0.00036, 0.00051]
Pima	LR	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	6.5e-16	-3e-33 [-3.4e-32, 3e-32]
Pima	LR	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	6.5e-16	-3e-33 [-3.5e-32, 3e-32]
Pima	RF	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	6.2e-16	-5.9e-33 [-3.7e-32, 2.5e-32]
Pima	RF	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	6.2e-16	-5.9e-33 [-3.7e-32, 2.6e-32]
Pima	XGBoost	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	4.4e-08	3.3e-25 [-2e-24, 2.5e-24]
Pima	XGBoost	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	4.4e-08	3.3e-25 [-1.9e-24, 2.5e-24]
Pima	MLP	$4\times\frac{1}{4}\times$ vs $1\times$	50	0.000 [0.000, 0.000]	5e-16	-3.9e-33 [-2.8e-32, 1.6e-32]
Pima	MLP	$4\times1\times$ vs $4\times$	50	0.000 [0.000, 0.000]	5e-16	-3.9e-33 [-2.6e-32, 1.7e-32]

Table 9: Equal-budget averaging: difference (averaged $-$ single run) in top-3 hit rate and relative L_2 error at the same total budget. "$4\times\frac{1}{4}\times$" is the mean of four runs at $\frac{1}{4}\times$ compared with one run at $1\times$; "$4\times1\times$" is four runs at $1\times$ against one at $4\times$. A positive Δ rel. L_2 means the average of runs is worse.

Dataset	Model	Contrast (A vs B)	n	Δ top-3 hit	Δ rel. L_2	Holm p (L_2)
German	LR	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	1.4e-15 [1.3e-15, 1.5e-15]	1.2e-13
German	LR	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	8.6e-16 [7.3e-16, 9.8e-16]	1.2e-12
German	LR	$16\times$ vs $4\times$	20	0.000 [0.000, 0.000]	-4.4e-16 [-6.5e-16, -2.3e-16]	0.024
German	LR	perm. vs kernel $1\times$	50	0.000 [0.000, 0.000]	-1.6e-15 [-1.8e-15, -1.4e-15]	1.2e-13
German	LR	perm. vs kernel $4\times$	50	0.000 [0.000, 0.000]	1.5e-15 [1.2e-15, 1.7e-15]	9e-13
German	LR	ℓ_1 off vs default $1\times$	50	0.077 [0.046, 0.112]	-0.22 [-0.24, -0.2]	1.2e-13
German	RF	$1\times$ vs $\frac{1}{4}\times$	50	0.007 [0.001, 0.014]	-0.014 [-0.016, -0.013]	1.2e-13
German	RF	$4\times$ vs $1\times$	50	0.007 [0.003, 0.013]	-0.0065 [-0.0072, -0.0059]	1.2e-13
German	RF	$16\times$ vs $4\times$	20	0.002 [0.000, 0.007]	-0.003 [-0.0035, -0.0026]	5.9e-05
German	RF	perm. vs kernel $1\times$	50	-0.008 [-0.015, -0.003]	0.0021 [0.0011, 0.0028]	5.6e-05
German	RF	perm. vs kernel $4\times$	50	-0.004 [-0.009, 0.000]	0.0017 [0.0014, 0.0021]	1.6e-09
German	RF	ℓ_1 off vs default $1\times$	50	0.030 [0.010, 0.054]	-0.11 [-0.13, -0.1]	1.2e-13
German	XGBoost	$1\times$ vs $\frac{1}{4}\times$	50	0.011 [0.005, 0.018]	-0.016 [-0.018, -0.014]	1.2e-13
German	XGBoost	$4\times$ vs $1\times$	50	0.004 [0.001, 0.008]	-0.0074 [-0.0082, -0.0068]	1.2e-13
German	XGBoost	$16\times$ vs $4\times$	20	0.000 [0.000, 0.000]	-0.0034 [-0.004, -0.0029]	5.9e-05
German	XGBoost	perm. vs kernel $1\times$	50	-0.002 [-0.005, 0.000]	-0.00042 [-0.0015, 0.00054]	1
German	XGBoost	perm. vs kernel $4\times$	50	-0.003 [-0.007, 0.000]	0.0005 [-4e-05, 0.00097]	0.18
German	XGBoost	ℓ_1 off vs default $1\times$	50	0.052 [0.024, 0.082]	-0.2 [-0.22, -0.18]	1.2e-13
German	MLP	$1\times$ vs $\frac{1}{4}\times$	20	0.022 [0.007, 0.038]	-0.024 [-0.027, -0.021]	5.9e-05
German	MLP	$4\times$ vs $1\times$	20	0.013 [0.003, 0.024]	-0.011 [-0.013, -0.0095]	5.9e-05
German	MLP	$16\times$ vs $4\times$	20	0.002 [0.000, 0.005]	-0.0049 [-0.0054, -0.0044]	5.9e-05
German	MLP	perm. vs kernel $1\times$	20	-0.007 [-0.018, 0.002]	0.0027 [0.00085, 0.0043]	0.14
German	MLP	perm. vs kernel $4\times$	20	-0.008 [-0.019, 0.000]	0.0026 [0.0019, 0.0033]	0.00048
German	MLP	ℓ_1 off vs default $1\times$	20	0.044 [0.012, 0.088]	-0.24 [-0.29, -0.19]	5.9e-05
ACS Inc.	LR	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	1.5e-15 [1.3e-15, 1.7e-15]	1.2e-13
ACS Inc.	LR	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
ACS Inc.	LR	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	2.3e-16 [4.3e-17, 4.3e-16]	1
ACS Inc.	RF	$1\times$ vs $\frac{1}{4}\times$	50	0.008 [0.001, 0.017]	-0.0098 [-0.011, -0.0088]	1.2e-13
ACS Inc.	RF	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
ACS Inc.	RF	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-2.4e-05 [-5.8e-05, 3.4e-16]	0.18
ACS Inc.	XGBoost	$1\times$ vs $\frac{1}{4}\times$	50	0.011 [0.002, 0.025]	-0.007 [-0.0077, -0.0063]	1.2e-13
ACS Inc.	XGBoost	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
ACS Inc.	XGBoost	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-9.3e-17 [-2.4e-16, 4.9e-17]	1
ACS Inc.	MLP	$1\times$ vs $\frac{1}{4}\times$	50	0.004 [0.000, 0.010]	-0.0065 [-0.0075, -0.0057]	1.2e-13
ACS Inc.	MLP	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
ACS Inc.	MLP	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	1.8e-16 [-5e-18, 3.8e-16]	1
ACS Emp.	LR	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	2.4e-15 [2.3e-15, 2.5e-15]	1.2e-13
ACS Emp.	LR	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	-9.4e-16 [-1.1e-15, -7.8e-16]	2e-12
ACS Emp.	LR	perm. vs kernel $1\times$	50	0.000 [0.000, 0.000]	-1.8e-15 [-1.9e-15, -1.6e-15]	1.2e-13
ACS Emp.	LR	perm. vs kernel $4\times$	50	0.000 [0.000, 0.000]	3.4e-15 [2.9e-15, 4e-15]	1.2e-13
ACS Emp.	LR	ℓ_1 off vs default $1\times$	50	0.012 [0.002, 0.028]	-0.061 [-0.074, -0.051]	1.2e-13
ACS Emp.	RF	$1\times$ vs $\frac{1}{4}\times$	50	0.005 [0.001, 0.011]	-0.0056 [-0.0062, -0.005]	1.2e-13
ACS Emp.	RF	$4\times$ vs $1\times$	50	0.001 [0.000, 0.002]	-0.0025 [-0.0028, -0.0022]	1.2e-13
ACS Emp.	RF	perm. vs kernel $1\times$	50	-0.005 [-0.012, -0.001]	0.0034 [0.0031, 0.0037]	1.2e-13
ACS Emp.	RF	perm. vs kernel $4\times$	50	-0.002 [-0.006, 0.000]	0.0023 [0.0021, 0.0025]	1.2e-13
ACS Emp.	RF	ℓ_1 off vs default $1\times$	50	-0.000 [-0.001, 0.000]	-0.013 [-0.016, -0.01]	1.2e-13
ACS Emp.	XGBoost	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	-0.0058 [-0.0064, -0.0052]	1.2e-13
ACS Emp.	XGBoost	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	-0.0026 [-0.0029, -0.0024]	1.2e-13
ACS Emp.	XGBoost	perm. vs kernel $1\times$	50	0.000 [0.000, 0.000]	0.0037 [0.0034, 0.004]	1.2e-13
ACS Emp.	XGBoost	perm. vs kernel $4\times$	50	0.000 [0.000, 0.000]	0.0024 [0.0022, 0.0026]	1.2e-13
ACS Emp.	XGBoost	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-0.02 [-0.023, -0.017]	1.2e-13
ACS Emp.	MLP	$1\times$ vs $\frac{1}{4}\times$	50	0.003 [0.000, 0.008]	-0.0053 [-0.0059, -0.0047]	1.2e-13
ACS Emp.	MLP	$4\times$ vs $1\times$	50	0.000 [0.000, 0.001]	-0.0022 [-0.0024, -0.002]	1.2e-13
ACS Emp.	MLP	perm. vs kernel $1\times$	50	-0.002 [-0.006, 0.000]	0.0034 [0.0031, 0.0038]	1.2e-13
ACS Emp.	MLP	perm. vs kernel $4\times$	50	-0.001 [-0.002, 0.000]	0.0022 [0.0019, 0.0024]	1.2e-13
ACS Emp.	MLP	ℓ_1 off vs default $1\times$	50	0.015 [0.000, 0.035]	-0.071 [-0.089, -0.057]	1.2e-13
Pima	LR	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	LR	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	LR	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-3.8e-18 [-9.3e-17, 9.2e-17]	1
Pima	RF	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	RF	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1

Dataset	Model	Contrast (A vs B)	n	Δ top-3 hit	Δ rel. L_2	Holm p (L_2)
Pima	RF	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-7.9e-17 [-1.4e-16, -1.9e-17]	0.2
Pima	XGBoost	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	XGBoost	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	XGBoost	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-2.4e-07 [-7.1e-07, 1.3e-16]	1
Pima	MLP	$1\times$ vs $\frac{1}{4}\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	MLP	$4\times$ vs $1\times$	50	0.000 [0.000, 0.000]	0 [0, 0]	1
Pima	MLP	ℓ_1 off vs default $1\times$	50	0.000 [0.000, 0.000]	-1.2e-16 [-2e-16, -4.7e-17]	0.076

Table 10: Planned paired contrasts per cell (A vs B; Δ is A minus B): budget steps, permutation versus Kernel SHAP at the same budget, and ℓ_1 off versus the library default at $1\times$. Holm-adjusted p for the relative-L_2 contrast.

Dataset	Model	Factor	n	$L \times S$	expl. SD (%)	factor SD (%) [CI]	var. ratio [CI]
German	LR	background	20	10×3	$< 10^{-9}$	15.0 [13.7, 16.4]	undefined (exact)
German	LR	refit	20	10×3	$< 10^{-9}$	0.0 [0.0, 0.0]	undefined (exact)
German	RF	background	20	10×3	1.2	15.2 [13.8, 16.7]	152 [104, 225]
German	RF	refit	20	10×3	1.2	16.1 [14.9, 17.5]	153 [105, 227]
German	XGBoost	background	20	10×3	1.4	17.5 [15.8, 19.6]	144 [100, 222]
German	XGBoost	refit	20	10×3	1.5	17.9 [16.7, 19.4]	133 [94, 185]
German	MLP	background	20	10×3	2	19.1 [17.4, 20.8]	81 [61, 109]
German	MLP	refit	20	10×3	2.2	44.4 [40.7, 48.6]	391 [331, 459]
ACS Emp.	LR	background	20	10×3	$< 10^{-9}$	14.9 [11.7, 19.1]	undefined (exact)
ACS Emp.	LR	refit	20	10×3	$< 10^{-9}$	0.0 [0.0, 0.0]	undefined (exact)
ACS Emp.	RF	background	20	10×3	0.66	22.0 [17.9, 26.4]	843 [464, 1842]
ACS Emp.	RF	refit	20	10×3	0.41	6.9 [6.1, 7.8]	242 [199, 327]
ACS Emp.	XGBoost	background	20	10×3	0.56	20.3 [16.3, 25.9]	1088 [719, 1878]
ACS Emp.	XGBoost	refit	20	10×3	0.39	4.7 [4.1, 5.4]	134 [116, 177]
ACS Emp.	MLP	background	20	10×3	0.57	20.4 [15.6, 26.8]	1102 [772, 1640]
ACS Emp.	MLP	refit	20	10×3	0.39	24.4 [21.2, 28.7]	4465 [3142, 6360]

Table 11: N4 probe: explainer-run SD, SD induced by redrawing the background (L levels $\times$ S seeds) and the ratio of the two variances, with bootstrap intervals. The ratio is descriptive for $m = 50$ at the $1\times$ budget and is undefined where the explainer is exact.

A.7 Value function

Table 12 compares the explained scalar: log-odds against predicted probability, same models, instances, background and seeds (ACS Employment, $1\times$). Reliability stays high with the probability (top-3 hit 0.995–0.997 against 1.000–1.000; Spearman 0.987–0.993 against 0.995–0.999), with a somewhat larger relative error (0.008–0.009 against about 0.004), as expected, since the probability is a nonlinear transform of the log-odds, and the additive LR is no longer exactly additive in this scale.

A.8 Use of LLMs and compute

The study (design, implementation, experiments, analysis and writing of this report) was carried out by LLM-based agents (Claude models) inside an automated research environment, working from a written task specification, with separate agents responsible for planning, experiments, figures, venue formatting and review. All numbers in the text come from result files produced by the released code; they enter the text through a generated macro file. Total compute was about 26 CPU-hours on a shared, oversubscribed CPU host (no GPU).

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Table 6: Stricter notions at $1\times$ (Kernel SHAP, ℓ_1 off): mean with nested-bootstrap 95% interval; n instances (German MLP: 20).

Dataset	Model	n	top-1	top-3 set	top-5 set	top-3 ordered	all signs
German	LR	50	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
German	RF	50	0.993 [0.979, 1.000]	0.990 [0.981, 0.997]	0.994 [0.988, 0.998]	0.959 [0.927, 0.983]	0.987 [0.983, 0.991]
German	XGBoost	50	1.000 [1.000, 1.000]	0.996 [0.990, 1.000]	0.987 [0.978, 0.994]	0.973 [0.949, 0.992]	0.988 [0.983, 0.993]
German	MLP	20	0.998 [0.988, 1.000]	0.977 [0.953, 0.995]	0.992 [0.982, 0.999]	0.927 [0.857, 0.983]	0.995 [0.991, 0.998]
ACS Emp.	LR	50	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
ACS Emp.	RF	50	1.000 [1.000, 1.000]	0.999 [0.998, 1.000]	0.997 [0.993, 1.000]	0.995 [0.986, 1.000]	0.974 [0.969, 0.979]
ACS Emp.	XGBoost	50	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.996 [0.988, 1.000]	1.000 [1.000, 1.000]	0.932 [0.930, 0.935]
ACS Emp.	MLP	50	1.000 [1.000, 1.000]	1.000 [0.999, 1.000]	0.998 [0.995, 1.000]	0.993 [0.978, 1.000]	0.995 [0.991, 0.997]

Table 12: Value-function sensitivity on ACS Employment at $1\times$: log-odds versus probability, mean over 50 instances with 95% bootstrap intervals.

Dataset	Model	Metric	log-odds [95% CI]	probability [95% CI]
ACS Employment	LR	Spearman $\lvert\phi\rvert$	1 [1, 1]	0.997 [0.996, 0.998]
ACS Employment	LR	top-3 Jaccard	1 [1, 1]	0.995 [0.984, 1]
ACS Employment	LR	rel. L_2 error	3.4e-15 [3.24e-15, 3.57e-15]	0.00525 [0.00465, 0.00585]
ACS Employment	XGBoost	Spearman $\lvert\phi\rvert$	0.995 [0.993, 0.996]	0.987 [0.983, 0.991]
ACS Employment	XGBoost	top-3 Jaccard	1 [1, 1]	0.988 [0.971, 1]
ACS Employment	XGBoost	rel. L_2 error	0.00417 [0.00377, 0.00458]	0.00885 [0.0079, 0.00983]
ACS Employment	MLP	Spearman $\lvert\phi\rvert$	0.999 [0.998, 0.999]	0.993 [0.989, 0.996]
ACS Employment	MLP	top-3 Jaccard	0.999 [0.995, 1]	0.993 [0.98, 1]
ACS Employment	MLP	rel. L_2 error	0.00351 [0.00317, 0.00389]	0.00825 [0.00722, 0.00952]

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